

Marshall Urias | Curriculum Vitae

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Current Affiliation

Assistant Professor of Economics — Peking University HSBC Business School

August 2018 – Present

Education

Ph.D. Economics — University of California, Irvine

August 2013 – May 2018

M.A. Economics — University of California, Irvine

August 2013 – December 2014

B.A. Economics, Minor in Math, magna cum laude — California State University, Fullerton

August 2008 – May 2013

B.A. Finance/Accounting, magna cum laude — California State University, Fullerton

August 2008 – May 2013

Fields of Interest

Macroeconomics, Monetary Theory, International Macroeconomics, International Trade

Papers

Productive demand, sectoral co-movement, and total capacity utilization, joint with Mario Silva. Economic Modelling (2025).

This paper develops a two-sector business cycle model in which aggregate demand affects total factor productivity and business formation. We estimate the model by Bayesian means using a wide array of observable series, included capacity utilization and sectoral labor hours, to decompose the role of goods market frictions, technology, and variable capital utilization on business cycle fluctuations. We find that demand shocks, especially shopping shocks, play a dominant role in explaining variation of investment, output, and the Solow residual over the next several years. Multifactor technology shocks are the most important among technology shocks and explain an increasing share of business cycle fluctuations at longer time horizons.

Sunk vacancy costs, endogenous product variety, and labor market frictions, joint with Mario Silva and Zhesheng Qiu

We develop a unified model of endogenous business formation and exit and study its implications for unemployment and vacancy dynamics. Heterogeneous, monopolistically competitive producers use labor and face exit risk from product obsolescence and idiosyncratic productivity shocks. The creation of both businesses and vacancies require sunk costs, making them partially predetermined: if a product line survives, unfilled vacancies are reposted. We estimate the model via Bayesian simulated method of moments using aggregate technology, product destruction, and match separation shocks. When a firm exits, it immediately eliminates jobs and posted vacancies and reduces product variety, so vacancy persistence closely tracks product-line longevity. In contrast, match separation shocks destroy only filled jobs and leave product lines intact, implying that only destruction shocks generate a Beveridge-curve comovement. The loss of variety also dampens job creation and employment via an aggregate demand externality. Adverse technology shocks prune product lines below a profitability cutoff amplifying labor-market effects. Quantitatively, the model matches the volatility, autocorrelation, and cross-correlation of unemployment, vacancies, business formation, employment-to-unemployment transitions, and establishment exit, with technology and destruction shocks playing the predominant roles.

Trade Intermediation

Exporters must decide which markets to sell to and how to deliver their products there. Alongside the conventional option of direct export, this model introduces an additional indirect export channel: intermediation. Intermediation is modeled as a Pissarides (2000) matching market which is then embedded within a standard intra-industry model of trade a la Melitz (2003). Firms' choice of which export channel to use will depend on the variety being sold, the destination, and ease of finding a trade

intermediary. Firms endogenously select into export channels such that high productivity firms export directly, moderate productivity firms export through intermediaries, and low productivity firms do not export at all. The model is able to generate several stylized facts that have been observed in empirical studies and offers tractable analytic explanations.

An Integrated Theory of Intermediation and Payments

This paper develops an integrated theory of intermediation and payments in wholesale and retail goods markets. The model synthesizes the search-theoretic approach to intermediation with the New Monetarist approach to payments. I consider two margins of intermediation, inventory and entry, within pure credit and pure currency markets. In a pure credit economy, the equilibrium is generically inefficient due to an inventory holdup problem and search externalities. Improving the bargaining position of intermediaries increases consumption and entry. In a pure currency economy, there is a two-sided holdup problem associated with middlemen's inventory choice and consumers' portfolio choice. This results in multiple steady state equilibria and a non-monotone response of consumption and entry to fundamentals. There exists a threshold nominal interest rate below which monetary policy is ineffective.

A Note on Liquidity and Unemployment

This note is intended to provide additional insight to the link between goods and labor markets in a New Monetarist model of liquidity as presented in Rocheteau and Nosal 2017 (RN). The framework integrates a model of money and credit into a Mortensen-Pissarides labor market to study the relationship between the availability of credit, firm entry, and unemployment. First, I show that even with a uniquely determined monetary equilibrium, there exists a non-monotone relationship between credit and unemployment dependent on the value of money. Second, I show that Nash bargaining reverses the response of unemployment to credit observed by RN. Specifically, more access to credit can decrease unemployment at the high equilibrium under Nash, whereas there is an increase in unemployment under proportional bargaining. The modeler's choice of bargaining protocol is not innocuous.

The Impact of Fuel Switching on EUA Prices

We estimate how daily carbon emission allowance (EUA) prices are affected by fuel switching costs and aggregate demand. Given that the data in the energy market is notoriously non-normal, we employ both OLS and quantile regression to measure the impact of switching prices over the entire distribution of EUA prices. We find that switching price has strong explanatory power, and moreover, varies significantly over the distribution of EUAs. Switching price has a significantly higher impact on EUAs that OLS regression predicts along the middle distribution of EUA prices. We also find significant positive impacts on EUA prices from aggregate demand measured by the Euro Stoxx50.

Investigating the Crowding Out Effect of Mainland Chinese Tourists on other International Tourists

We measure the crowding out effect of mainland Chinese tourists on tourists of other nationalities on six tourist recipient locations: Macau, Japan, Korea, Singapore, Malaysia, and Taiwan. We show that a 1 percent increase in Chinese tourist arrivals results in a 0.182 percent decrease in arrivals from other nationalities. We then examine the impact of tourist arrivals on the local price level of tourist goods and show that a 1 percent increase in Chinese tourists leading to an average 0.097 percent increase in the local price index.

Ongoing Projects

Middlemen and Exchange Rates

Building on search theoretic models of intermediation, this paper shows how brokers and dealers in foreign exchange markets can impact the degree of exchange rate passthrough in open economies. The model shows purchasing power parity obtains as a result of consumers currency portfolio choices and random matching of consumers and sellers in a goods market. In the absence of forex brokers, the degree of exchange rate passthrough depends on the production technology of sellers. In particular, so long as the technology exhibits convex costs, relative PPP fails and exchange rate passthrough is incomplete. In the presence of forex dealers, there exists incomplete exchange rate passthrough and failure of relative PPP, even if passthrough is complete.

The Effect of Banking Credit on the Non-oil Current Account: Evidence from Azerbaijan

This paper uses an ARDL model to empirically estimate the short-run and long-run dynamics of various components of bank credit on the current account of Azerbaijan. Specifically, we decompose bank credit into four dimensions: public, private, business, and household. We find that the current account deteriorates both in the long-run and short-run as total bank credit expands; this is primarily driven by the growth of business credit.

TFP Spillovers Along the Belt and Road

This research examines the impact of Chinese import penetration on the total factor productivity (TFP) of Belt and Road countries. We utilize two different approaches to measure TFP: the conventional Cobb-Douglas production function and a non-parametric Data Envelopment Analysis (DEA) with a Malmquist index to dissect TFP growth into technical change (the process of shifting outward an economy's technology frontier through disruptive innovation) and efficiency change (the process of moving closer to an existing technology frontier through maximizing with a given set of inputs). A two-stage least squares (2SLS)

estimator with fixed effects is applied to a panel dataset of 66 BRI member states for the years 2002 until 2019. The production function approach shows a positive effect of import penetration on the TFP of BRI countries for consumer goods, but no effect on capital and intermediate goods. On the other hand, when using a non-parametric Malmquist-style measure of TFP, a negative overall effect on TFP growth is found. The decomposition of productivity growth into EC and TC shows that this effect is driven by a large negative impact on innovation change (TC), which is partly attenuated by a positive effect on efficiency change (EC). This suggests that Chinese imports, albeit helping countries to move closer to their existing technology frontier, negatively affect the innovativeness of BRI countries, and with it, the overall factor productivity as a whole.

Measuring the Welfare Cost of Inflation in China

This paper seeks to measure the welfare cost of inflation for the Chinese economy following the methodology used by Lucas (2000) and rationalize these estimates from new advances in monetary economics identified in Craig and Rocheteau (2008). The aim of this project is to estimate the “welfare triangle” from an estimated money demand curve for China, explore how sensitive these estimates are to interpreting the money demand function within a structural model, and evaluating whether or not the Friedman rule is optimal.

Capital Controls and Firm Financing

This project is based off of work done with my advisee Jim Bakker. The project builds a structural model of the Chinese economy to include financial repression (preferential loan treatment of SOEs), capital controls, and firms financing decisions under a competitive banking sector and onshore/offshore bond markets. The model generates passthrough from the short term deposit rate to the financing cost of new capital providing a link between the financial sector and the real economy. The degree to which capital controls can lower the average financing cost depends on the degree to which the government favors SOE bank loans, and the savings preferences of the economy. Thus, there exist policy trade-offs between achieving lower financing costs and stimulating POE investment, versus stimulating SOE investment.

SOEs, POEs and Interest Rate Passthrough

This project is motivated by two ideas: (i) State owned enterprises (SOEs) in China have much easier access to bank credit than privately owned enterprises (POEs), so that POEs must rely more on internal liquidity to finance capital expenditures, and (ii) passthrough from short term nominal interest rates to real interest rates is greater when firms have better access to external liquidity. This project includes these two different types of ownership structures in a search and bargaining model of firm credit that has both internal liquidity (retained earnings) and external liquidity (bank credit) to examine how the ownership structure of China’s economy affects the ability of monetary policy to affect real interest rates.

Journals Served as Referee

Economic Modeling, Economic Inquiry

Teaching Experience

Instructor — Peking University HSBC Business School

August 2018 - Present

Money and Banking: This course focuses on monetary theory, financial intermediation, and central bank policy. The first part of the curriculum develops a theoretical framework used to explain the existence of money and banks, demonstrating key relationships between the availability of liquid assets, inflation, output, and regulations imposed on banks. Second, we discuss the real world business of banking which includes deposit creation, fractional reserve banking, and the interbank market. Third, we discuss the role of central banks in modern economies and analyse the conduct of monetary policy. [9 quarters]

International Trade: This course discusses the canonical models of international trade and international macroeconomics. The first part of the curriculum introduces students the foundational theories of trade including Ricardio, Heckscher-Ohlin, and models of monopolistic competition. The second part develops a RBC model of an open economy based on the intertemporal approach to the current account. [6 quarters]

Introduction to International Trade: This course is an undergraduate level introduction to theory and applications of international trade. We discuss the foundational models including Ricardo, Heckscher-Ohlin, specific factors, and monopolistic competition. This course also introduces students to procuring data related to international flows of goods, services, and assets and includes a discussion of trade policy with a focus on tariffs. [2 quarters]

Macroeconomic Analysis: This course discusses the search and bargaining approach to understanding economics characterized by frictions in goods, labor and financial markets. In departing from classical equilibrium theory, this class explores phenomena such as unemployment, pricing strategies, production capacity choices, financial contracts, and business cycles. [3 quarter]

Macroeconomics: This course discusses foundational models of macroeconomics at an advanced undergraduate or early graduate level. Topics include the Solow model, Diamond-Mortensen-Pissardes labor search, theories of the price level and nominal interest rates, and finally a IS-LM-MP model of short-run. [1 quarter]

Microeconomics I (Quant Finance): This course introduces finance students to the fundamentals of microeconomics including consumer behavior, producer behavior, equilibrium, and classic oligopoly structures. [1 quarter]

Instructor — California State University, Fullerton

August 2016 – May 2018

Intermediate Macroeconomic Analysis: Introduces students to the determinants of national output, prices, interest rates, business cycle fluctuations, employment and economic growth. Also considers the theory of monetary and fiscal stabilization policies. [3 semesters]

Teaching Assistant — University of California, Irvine

August 2013 – May 2018

Basic Economics (II): Introduces students to the fundamentals of macroeconomics. Curriculum includes monetary and fiscal policy, inflation, and unemployment. [1 quarter]

Intermediate Economics (I, II, III): Three course core sequence for economics majors. Includes determinants of demand and supply; market structures, market imperfections; explanations of unemployment, inflation, growth, and business cycles. [8 quarters]

Global Economy: Acquaints students with the fundamental patterns of the global economy. Introduces fundamental economic concepts with an emphasis on international trade, globalization, and international finance. [1 quarter]

International Money: Familiarizes students with open economy macroeconomics and determination of exchange rates. Curriculum includes asset-market approach to the balance of payments; macroeconomic policies under fixed and floating exchange rates; the international monetary system and institutions. [2 quarters]

Guest Lecturer — University of California, Irvine

August 2017

Special Topics in Monetary Theory: Summer school for 2nd year Ph.D. students introducing New Monetarist models on the coexistence of money and credit. Exposit pure credit economies with endogenous borrowing limits and economies in which money is essential and coexists with credit. [1 lecture]

Certificate of Teaching Excellence Program (CTEP) — University of California, Irvine

March 2016 - Present

The Center for Engaged Instruction provides extended pedagogical training for UCI graduate students. Participants are able to design lessons using inclusive, high-impact, best-practices, and evidence-based pedagogical principles; analyze and assess teaching practices; and effectively present and facilitate learning.

Economics Learning Center (ELC) — University of California, Irvine

September – December 2015

The ELC is a tutoring resource that offers free walk-in tutoring for UCI students enrolled in certain courses. I facilitated discussion and collaboration among students that use the center. Tutoring is usually done one-on-one or in small groups.

Student Reviews Available upon Request

Experience as Advisor

Li Xinda, M.A. Finance

Thesis: Corruption and Innovation: Evidence from the Chinese Market

Henry Yu, M.A. Finance

Thesis: An Empirical Investigation on Industrial Structure Upgrading, Loan Risk and Bank Performance of Regional Commercial Banks

Jim Bakker, M.A. Economics

Thesis: Capital Controls and POE Funding Channels

Sven Wessling, M.A. Finance

Thesis: Risk Linkages Between Sovereigns and Banks

Leonard Wise, M.A. Finance

Thesis: The Stability of the Financial Determinants of Corporate Cash Holdings

Danny Bonfil, M.A. Economics

Thesis: MAD Agents: Low Stakes still Mistakes?

Philip Hackel, M.A Economics

Thesis: Chinese Exports and Total Factor Productivity: Evidence from Belt and Road Countries

Li Yiqian M.A. Finance

Thesis: The Impact of Foreign Investment on Total Factor Productivity

Chen Zhuo M.A. Finance

Thesis: The Impact of Fuel Switching on EUA Prices

Desmond Woo Chien Yang, M.A. Economics

Thesis: The Crowding Out Effect of Chinese Tourism

Jasper Lim, M.A. Finance

Thesis: The Effect of Monetary Policy Shocks on Bank Lending in China

Tim Vree, M.A. Economics

Thesis: Sunk v. Fixed Costs in a Labor Search Model: Evidence from Germany

Samuel Khaw Tse Yung, M.A. Finance

Thesis: Import Penetration and Firm Innovation: Evidence from China